

# Domain adaptation for *Catalan–Basque* machine translation via synthetic data and continued fine-tuning

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## 1 Motivation & research questions

Catalan–Basque (CA–EU) machine translation remains highly under-resourced, particularly in specialized domains. Existing multilingual models often struggle with this language pair, while high-quality in-domain parallel corpora are extremely limited.

**Core question:** Given the scarcity of domain-specific parallel corpora, can synthetic bilingual data generated from monolingual resources significantly improve CA–EU machine translation?

- Can pivot-based corpus construction and back-translation generate effective training data for low-resource specialized domains?
- Does domain-specific fine-tuning improve in-domain translation quality over the general model?
- Is continued fine-tuning from a general-domain checkpoint better than direct domain-specific SFT?

## 2 Two domains, two quality criteria

### LITERARY

Quality depends on fluency, creativity and stylistic naturalness.

### CLINICAL

Quality depends on terminological accuracy and clinical clarity.

## 3 Datasets

### SOURCE CORPORA

#### GENERAL

AINA CA–EU Parallel

100k pairs

#### LITERARY A

CTILC CA–ES

~30k pairs ES pivot → EU synthetic

#### LITERARY B

EhuHac ES–EU

100k pairs ES pivot → CA synthetic

#### CLINICAL

E3-RC EU Clinical

~1,300 pairs · ca→eu only

### TRAIN · VALIDATION · TEST SPLITS

#### GENERAL

eu→ca & ca→eu

90 / 5 / 5 % split

TRAIN

87.2k

43,661 ca→eu · 43,568 eu→ca

VALIDATION

4.8k

2,388 ca→eu · 2,458 eu→ca

TEST

4.8k

2,412 ca→eu · 2,435 eu→ca

#### LITERARY

eu→ca & ca→eu

avg 110 tok

TRAIN

102k

75,444 ca→eu · 26,900 eu→ca

VALIDATION

5.7k

4,217 ca→eu · 1,468 eu→ca

TEST

5.7k

4,261 ca→eu · 1,426 eu→ca

#### CLINICAL

ca→eu only

avg 1078 tok

TRAIN

1,174

VALIDATION

65

TEST

66

## 4 Synthetic data construction

LITERARY A: CA (CTILC)

CA monolingual corpus

CA → ES (pivot) → EU translate

ca-literary

LITERARY B: EU-ES (EHUHAC)

EU-ES corpus

ES → CA translate

eu-literary

CLINICAL EU-CA (E3C)

EU monolingual corpus

EU → CA

eu-clinical

**Round-trip technique:** for model fine-tuning, we reverse the translation direction, using machine-translated texts as the source language and original texts (high quality) as the target language.

## 5 Training setup

|                      |                                  |
|----------------------|----------------------------------|
| Fine-tuning model    | HiTZ/Latxa-Qwen3-VL-8B-Instruct  |
| Translation model    | HiTZ/Latxa-Llama-3.1-8B-Instruct |
| Adaptation           | LoRA · 4-bit NF4 (bitsandbytes)  |
| Max length           | 768 tokens                       |
| Epochs               | 3                                |
| Seed                 | 42                               |
| Batch                | 4                                |
| Grad accum           | 8                                |
| Learning rate        | 5e-5 · cosine schedule           |
| Checkpoint selection | Best BLEU on validation          |

**Literary domain:** each parallel pair → two instruction samples eu2ca + ca2eu .

**Clinical domain:** single direction (due to data limitations) ca2eu .

## 6 Model variants & experimental design

GENERAL — DIRECT SFT ON GENERAL-DOMAIN PARALLEL DATA

Latxa-Qwen3 base

SFT bidirectional

general

LITERARYV1 — DIRECT LITERARY SFT (SYNTHETIC PARALLEL DATA THROUGH PIVOTING)

Latxa-Qwen3 base

SFT bidirectional

literaryv1

LITERARYV2 — CONTINUED FROM GENERAL

general

continued literary SFT

literaryv2

CLINICALV1 — DIRECT CLINICAL SFT

Latxa-Qwen3 base

SFT CA-EU

clinicalv1

CLINICALV2 — CONTINUED FROM GENERAL

general

continued clinical SFT

clinicalv2

## 7 General domain results (bidirectional)

| MODEL     | DIR   | CHRF++↑ | BLEU↑ | TER↓  | COMET↑ |
|-----------|-------|---------|-------|-------|--------|
| Baseline  | eu→ca | 10.93   | 0.60  | 300.3 | 45.17  |
| Baseline  | ca→eu | 17.10   | 1.29  | 296.3 | 48.54  |
| generalv1 | eu→ca | 45.03   | 18.51 | 75.53 | 80.61  |
| generalv1 | ca→eu | 41.63   | 9.92  | 88.02 | 80.75  |

Baseline = Latxa-Qwen3-VL-8B-Instruct zero-shot.

## 8 Literary domain results (bidirectional)

| MODEL        | DIR   | CHRF++↑ | BLEU↑ | TER↓  | COMET↑ |
|--------------|-------|---------|-------|-------|--------|
| Baseline     | eu→ca | 11.96   | 0.82  | 151.3 | 41.55  |
| Baseline     | ca→eu | 16.59   | 0.63  | 266.6 | 44.82  |
| generalv1    | eu→ca | 27.97   | 4.96  | 89.44 | 66.68  |
| generalv1    | ca→eu | 29.58   | 2.82  | 101.8 | 67.76  |
| literaryv1 ★ | eu→ca | 36.13   | 8.96  | 85.08 | 69.61  |
| literaryv1 ★ | ca→eu | 26.90   | 2.44  | 99.60 | 65.29  |
| literaryv2   | eu→ca | 34.51   | 7.44  | 87.66 | 68.72  |
| literaryv2   | ca→eu | 25.87   | 2.31  | 100.7 | 64.34  |

★ Direct SFT (v1) beats continued adaptation (v2).

## 9 Clinical domain results (ca→eu)

| MODEL        | CHRF++↑ | BLEU↑ | TER↓   | COMET↑ |
|--------------|---------|-------|--------|--------|
| Baseline     | 11.03   | 0.64  | 104.3  | 50.56  |
| generalv1    | 10.60   | 0.38  | 98.10  | 56.56  |
| clinicalv1 ★ | 40.20   | 19.43 | 101.09 | 76.25  |
| clinicalv2   | 38.73   | 18.50 | 104.64 | 75.02  |

Clinicalv2 achieves best chrF++ & BLEU and lowest TER; clinicalv1 leads on COMET.

## 10 Token-matched results (ca→eu)

| MODEL        | CHRF++↑ | BLEU↑ | TER↓   | COMET↑ |
|--------------|---------|-------|--------|--------|
| clinicalv1 ★ | 40.20   | 19.43 | 101.09 | 76.25  |
| literaryv1   | 24.73   | 2.28  | 97.03  | 65.41  |

Each model is trained on the same amount of data (~360,000 tokens) and evaluated on their respective domains.

## 11 Key findings

- Domain-specific fine-tuning is the main driver of improvement, in particular for clinicalv1.
- Cross-domain transfer helps but is insufficient for adapting v2 models to dedicated domains.
- Literary: direct SFT outperforms continued adaptation (literaryv2) and zero-shot across all metrics.
- Generating training pairs either via a pivot language (ES) or direct translation successfully compensates for the absolute lack of parallel corpora.
- Clinical domains are more "learnable": under strict token-matching, clinical SFT yields for most metrics substantially higher results than literary SFT.

## 12 Limitations & future work

**Limitations:** Small clinical test set (N=66); single-directional clinical training; reliance on synthetic supervision.

### CORPUS

Due to high reliance on synthetic data, explore SFT results on human-translated parallel data. Extend clinical to eu2ca.

### ABLATIONS

Test all possible pivot language choices and SFT data sizes variation between domains. Try English prompting.

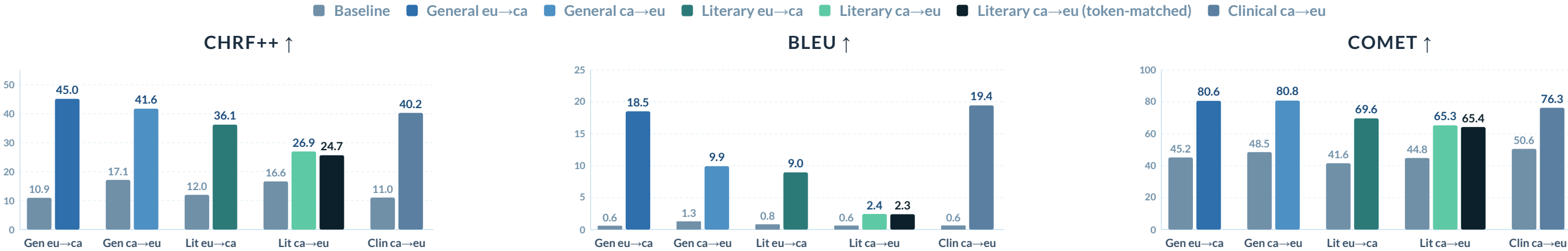
### EVALUATION

Incorporate human evaluation.

### SINGLE MODEL

Single-architecture setup limits generalization across model families.

## Best model per domain vs. zero-shot



### RESOURCES

projecte-aina/CA-EU\_Parallel\_Corpus · EhuHac EU literary corpus · CTILC CA literary corpus · E3C clinical EU corpus

### BASE MODEL (FINE-TUNING)

HiTZ/Latxa-Qwen3-VL-8B-Instruct

### TRANSLATION MODEL (DATA CONSTRUCTION)

HiTZ/Latxa-Llama-3.1-8B-Instruct

### GITHUB REPOSITORY

